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OpenShift Virtualization

High-level overview

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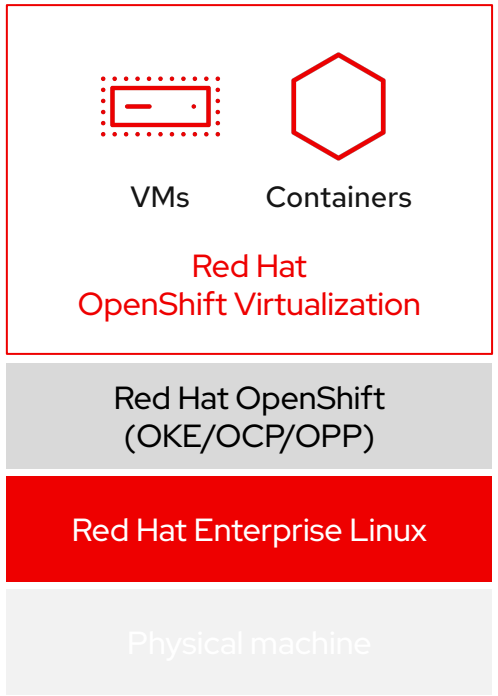
Agenda

- Context, value, and how it compares to classic stacks
- Architecture, networking, storage, and migration
- Security, GitOps automation, and day-two operations
- Use cases, takeaways, and where to learn more

Red Hat OpenShift Virtualization

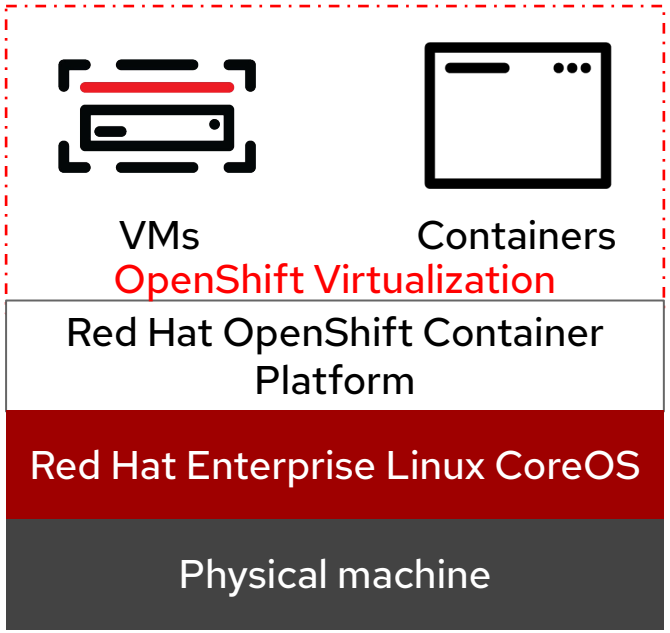
The modern option for general purpose virtualization

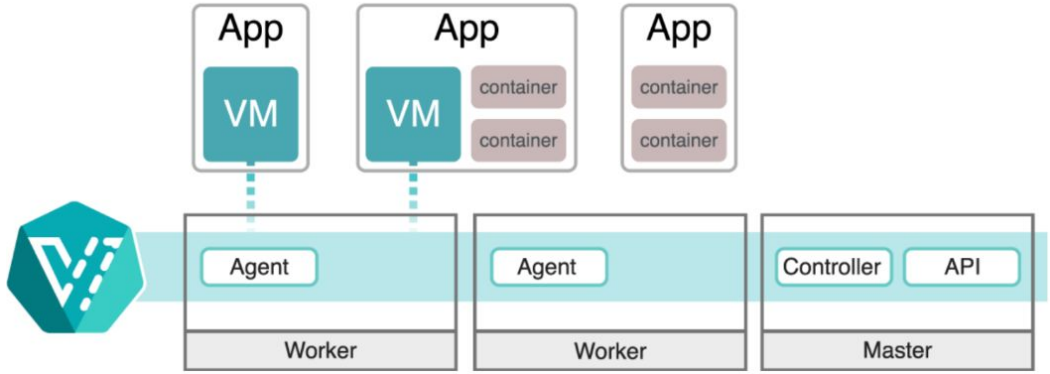


- **Unified platform**
  for virtual machines and containers
- **Consistent management**
  tools, interfaces, and APIs incl. ACM and AAP integrations
- **Performance and stability**
  of Linux, KVM, and qemu
- **Healthy open source community**
  the KubeVirt project is a top 10 CNCF active project, with 200+ contributing companies
- **Supports Microsoft Windows**
  guests through Microsoft SVVP
- **Inbound guest migration**
  using Ansible Automation Platform + Migration Toolkit for Virtualization, Training and Consulting
- **Diverse ecosystem**
  of Red Hat & partner operators

What is OpenShift Virtualization?

- Unified platform for running VMs and Containers
- Include features of the OpenShift application platform
- Run VMs in OpenShift
- Performance, stability, scalability, and reliability of KVM, the Linux kernel-based hypervisor
- Manageability and ecosystem of OpenShift
- Supports Microsoft Windows guests – Microsoft Server Virtualization Validation Program (SVVP)





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Hybrid cloud application platform

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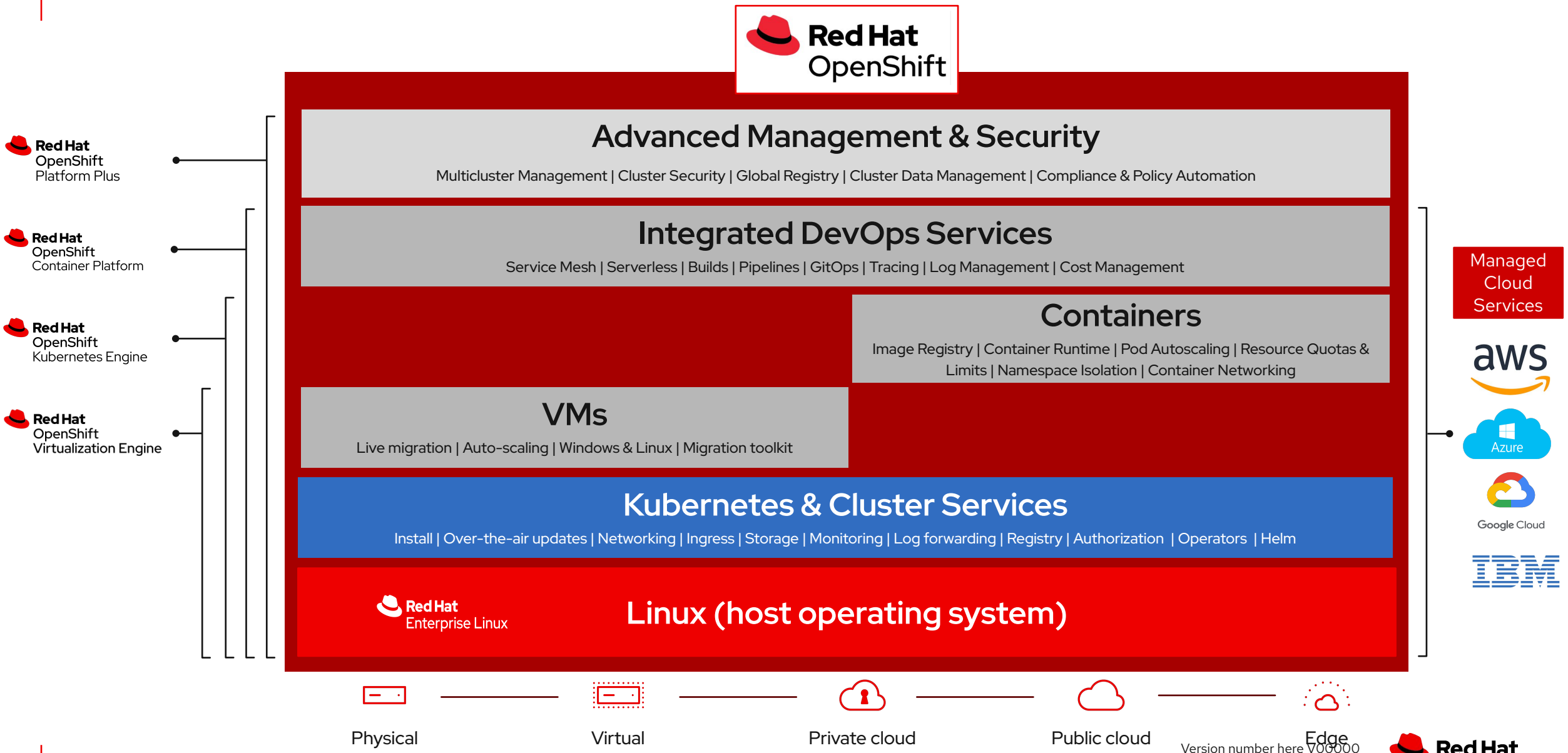

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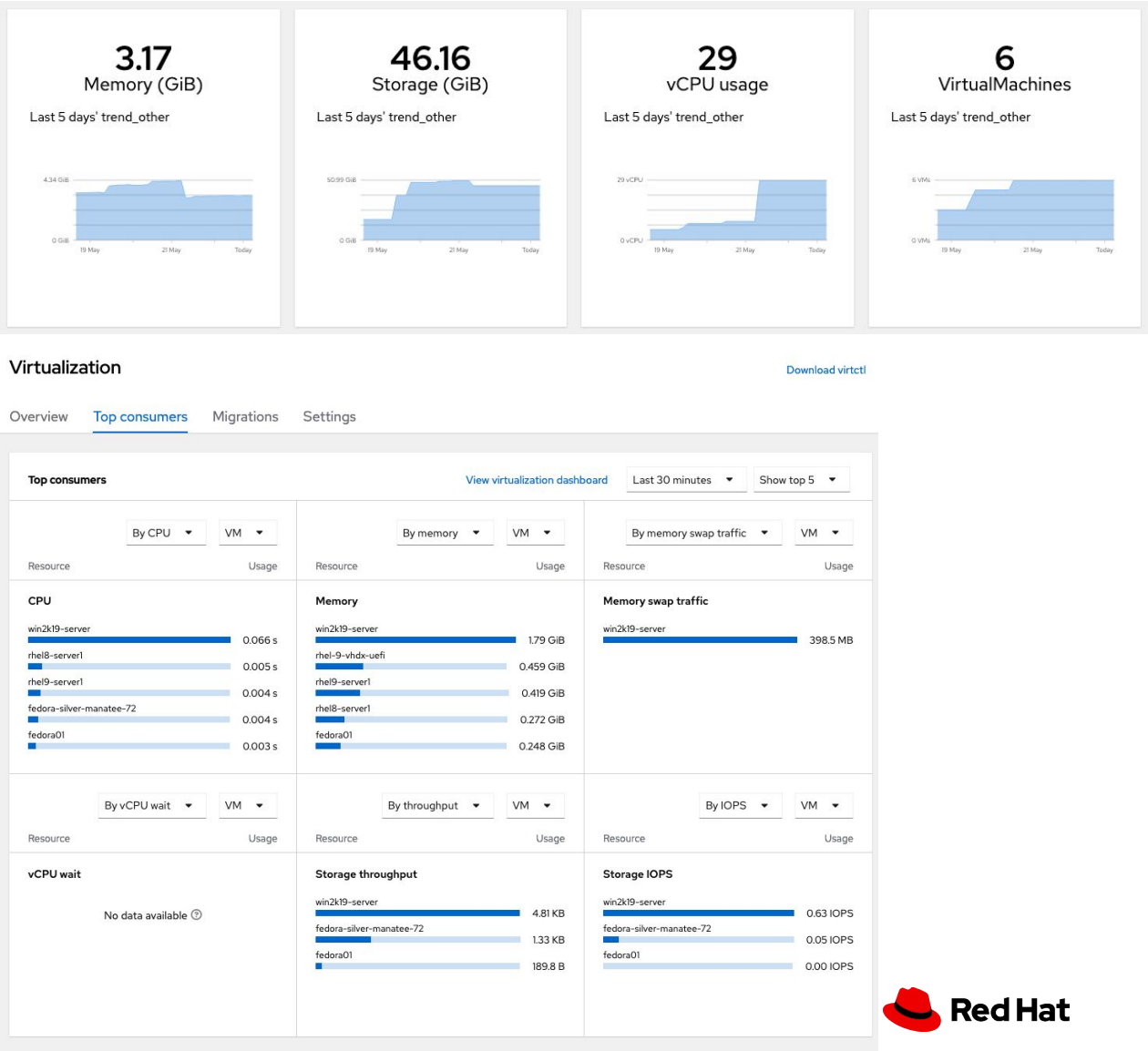
OpenShift Virtualization Overview

- Virtual Machine
  - Provisioning, Deprovisioning
  - Life Cycle Management
  - Live Migration (vMotion equivalent)
- Platform
  - Storage
    - Software Defined (Block, File, NFS, Object Storage)
    - Traditional Storage with Container Storage Interface
      - Dell, EMC, HPE, Hitachi, IBM, Pure etc
  - Network
    - Software Defined (OpenShift OVN)
    - Multiple Networks with VLANs segregation (Multus)
  - Load Balancing
    - MetalLB, F5 etc
- Backup and Restore
  - OADP (Valero), Kasten, Portworx, NetApp, Veritas etc
- Migration to OpenShift
  - vSphere, ESXi, OVA
  - Red Hat Virtualization, OpenStack
  - Hyper-V and other KVM variants (Automate with Ansible)



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De-Risking Your Virtualisation Technology Investme

New and modern applications will be built on containers. They provide new levels of agility and empowers organisations to accelerate their digital capabilities.

However, **not all applications** can or are **ready** to be containerized and operate in microservices.

In most organizations, the **journey** will be a **multi-phased approach**, requiring IT operations to maintain and **coexist** workloads with both virtual machines and containers in their IT landscape.

Future-Proof Your Virtualization Strategy

1. **Rehosting** by "shifting" virtual machine workloads into the OpenShift platform
2. **Replatform** by "upgrading" the application into a container-based architecture
3. **Refactor** applications from monolithic to microservices

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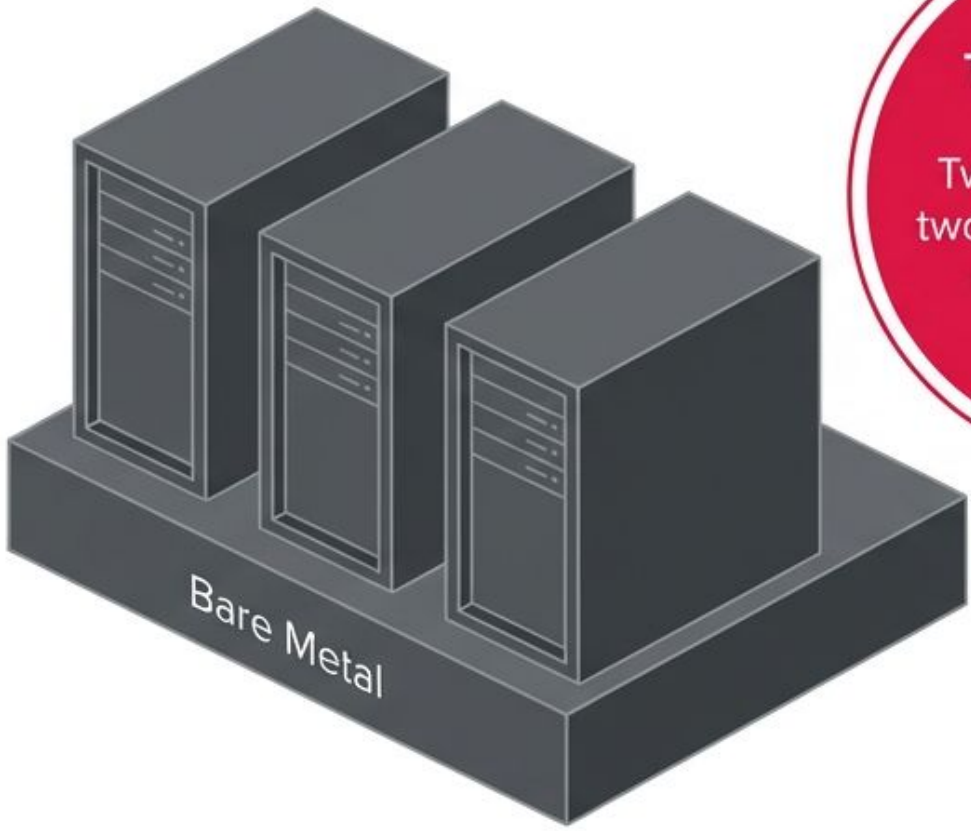
The VM Silo

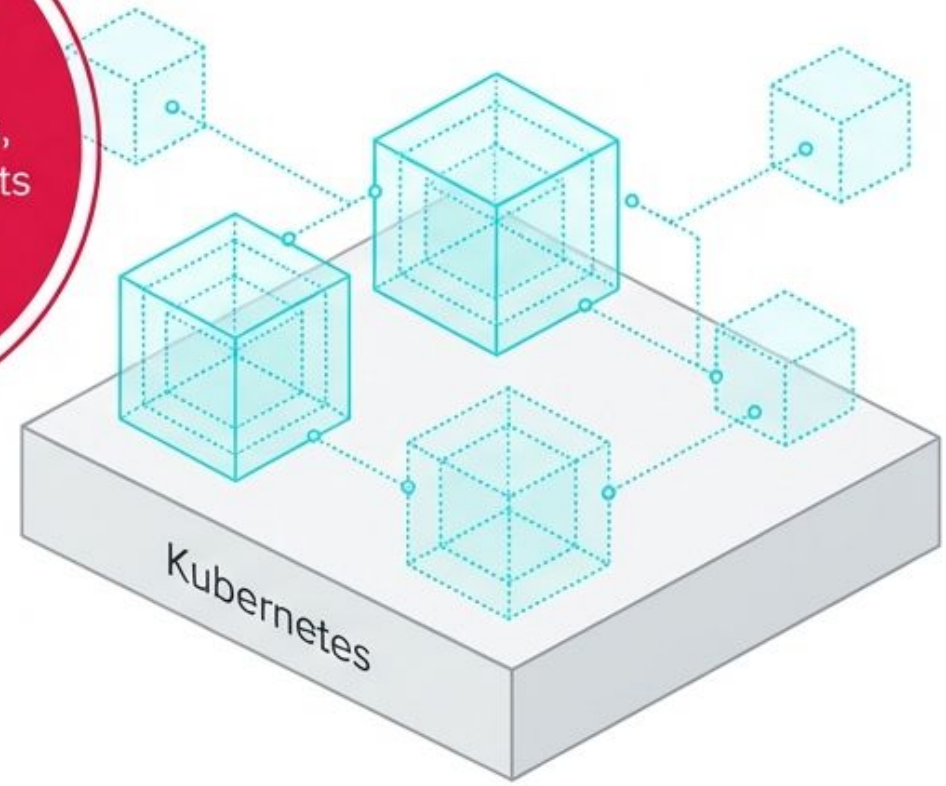
- Heavyweight, stateful workloads
- Dedicated hypervisors (vSphere, RHV)
- Isolated networking and storage operations



The Operational Divide:
Two control planes, two skillsets, two sets of infrastructure overhead.

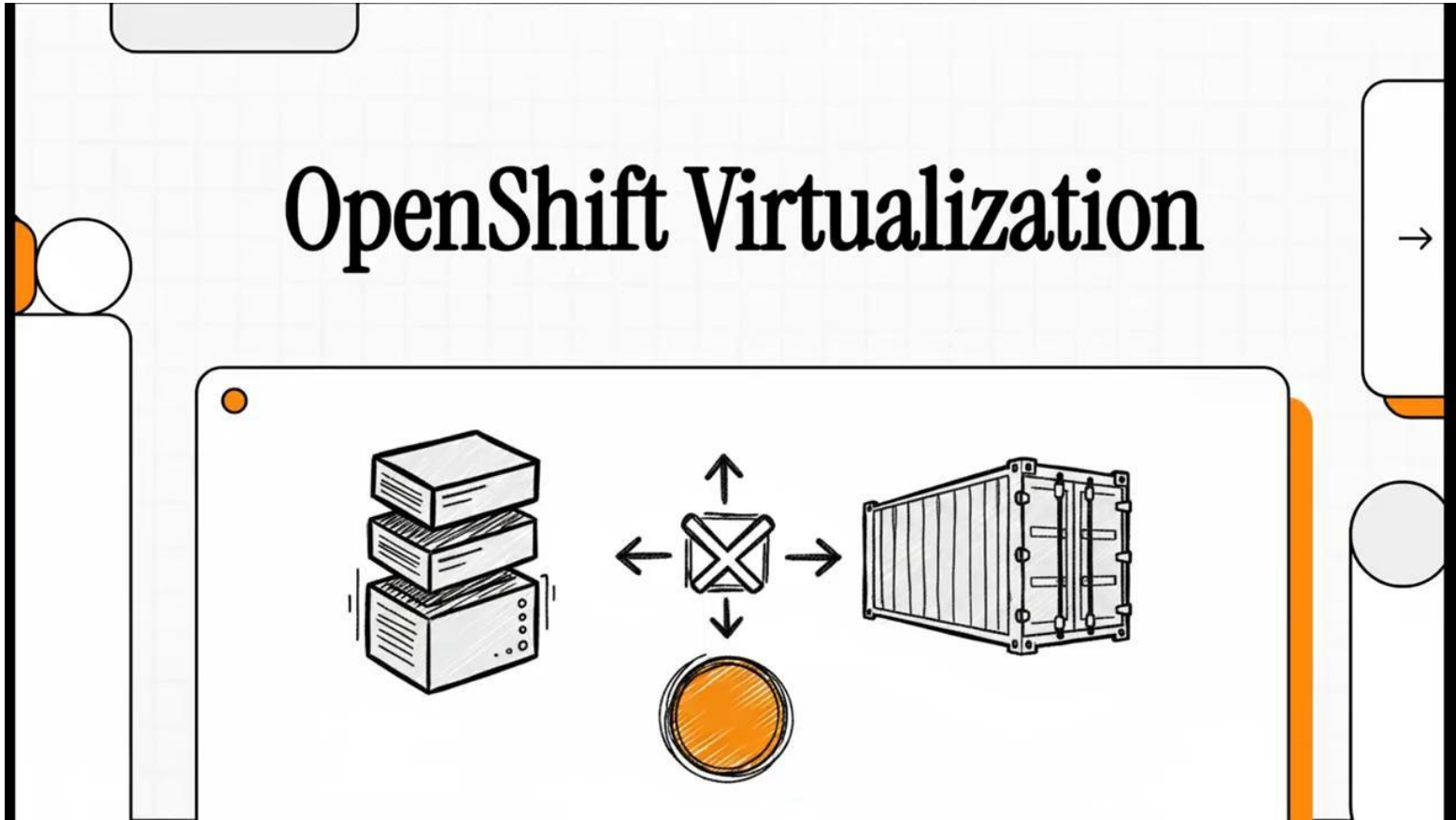
The Container Silo

- Ephemeral, stateless scaling
- Orchestrated by Kubernetes
- Declarative API management



NotebookLM

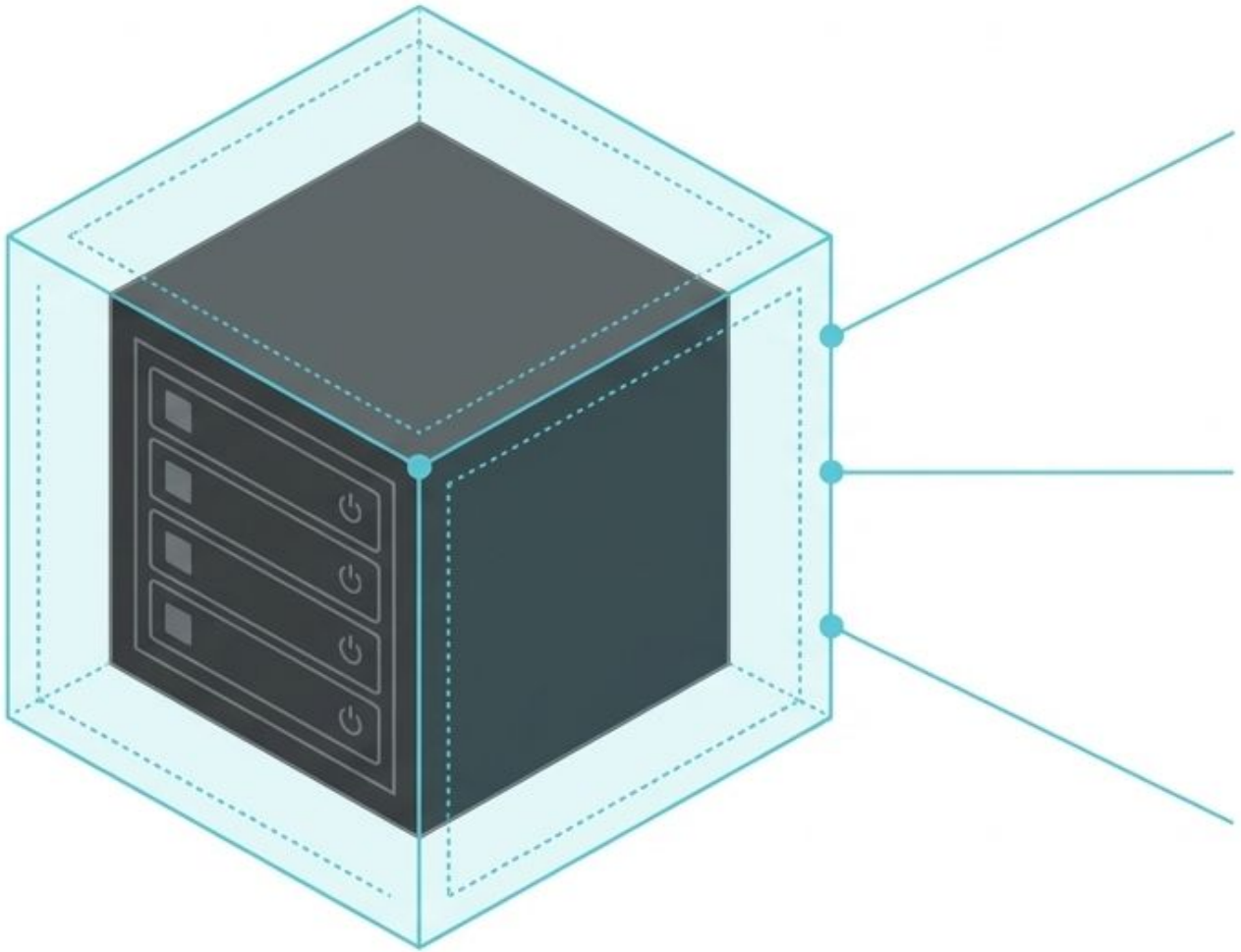
Foundation



What it is

- Runs virtual machines on OpenShift alongside containers
- Built on KubeVirt and Kubernetes-native APIs
- Managed like any other OpenShift workload
- Same cluster ops, networking, and storage patterns

The Paradigm Shift: VMs as Native Kubernetes Objects



No Hypervisor Segregation

VMs run alongside container workloads on the exact same physical nodes.

Declarative Management

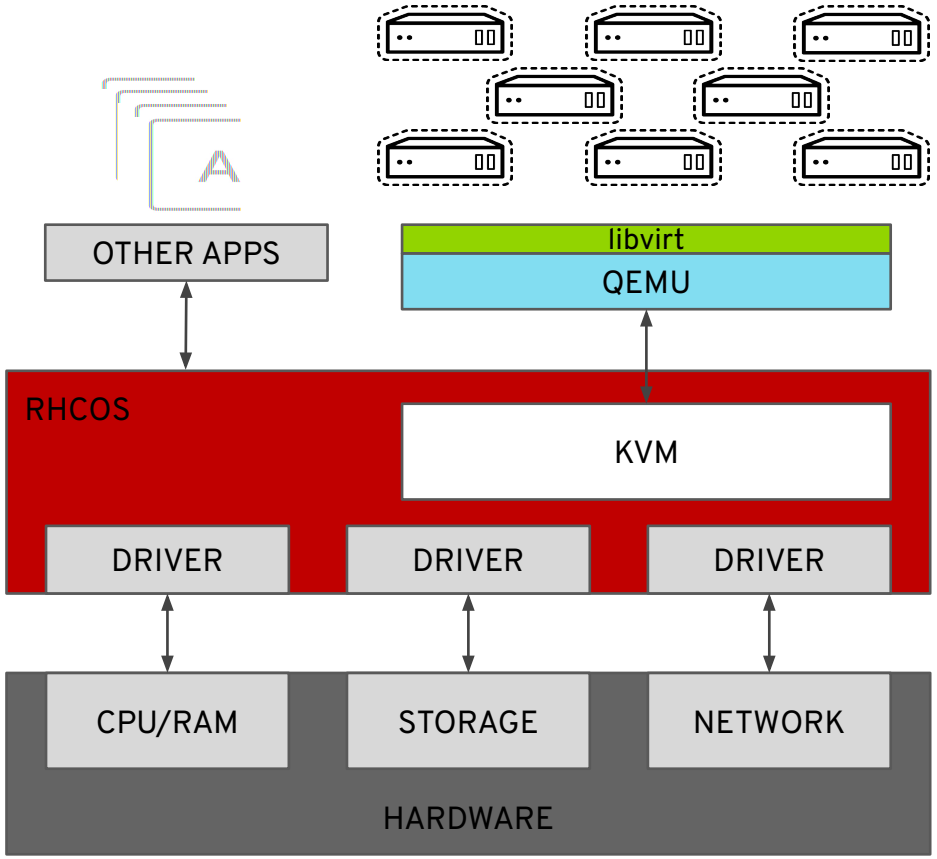
VMs are managed via Kubernetes Custom Resource Definitions (CRDs).

Unified Kubelet

The exact same primary node agent (kubelet) monitors the health of both microservices and virtual machines.

OpenShift Virtualization uses KVM

- OpenShift Virtualization uses KVM, the Linux kernel hypervisor
- KVM is a core component of the Red Hat Enterprise Linux kernel
  - KVM has 10+ years of production use: Red Hat Virtualization, Red Hat OpenStack Platform, and RHEL all leverage KVM, QEMU, and libvirt
- QEMU uses KVM to execute virtual machines
- `libvirt` provides a management abstraction layer
- Currently supported on x86 bare metal
- For other platforms contact Product Management for roadmap



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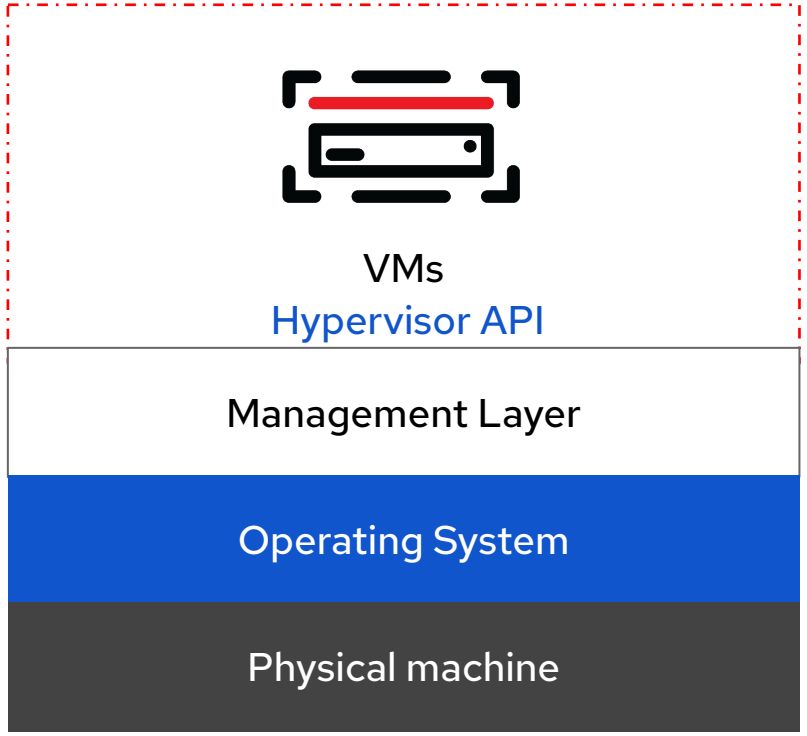
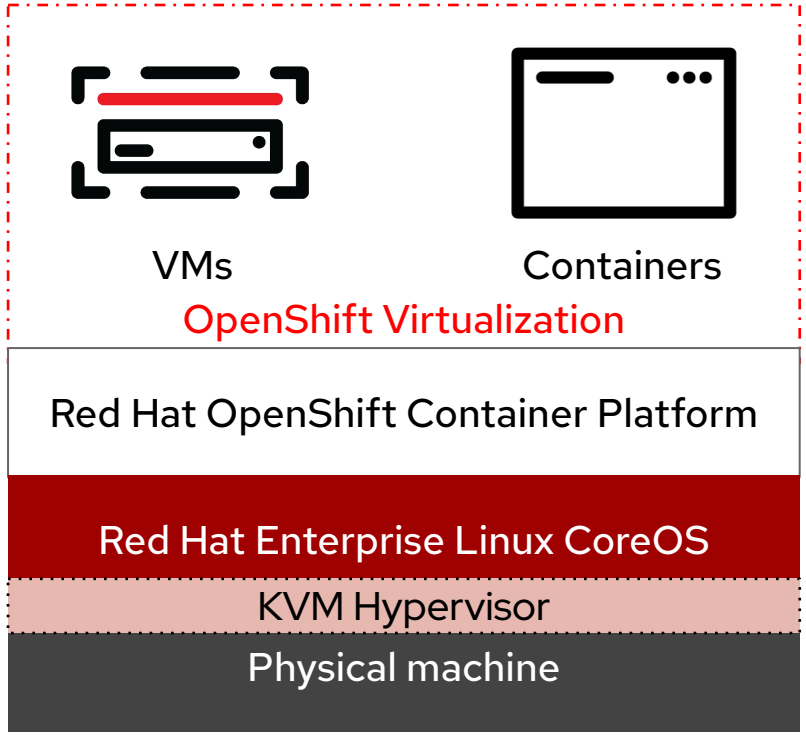
Traditional Type 1 Hypervisor vs OpenShift Virtualization





OpenStack vs RHV vs OpenShift Virtualization

VMs Nova Compute	VMs RHV Hypervisor	VMs Containers OpenShift Virtualization
Red Hat OpenStack (RH-OSP)	Red Hat Virtualization (RHV)	Red Hat OpenShift Container Platform
Red Hat Enterprise Linux	Red Hat Enterprise Linux	Red Hat Enterprise Linux CoreOS
KVM Hypervisor		
Physical machine	Physical machine	Physical machine

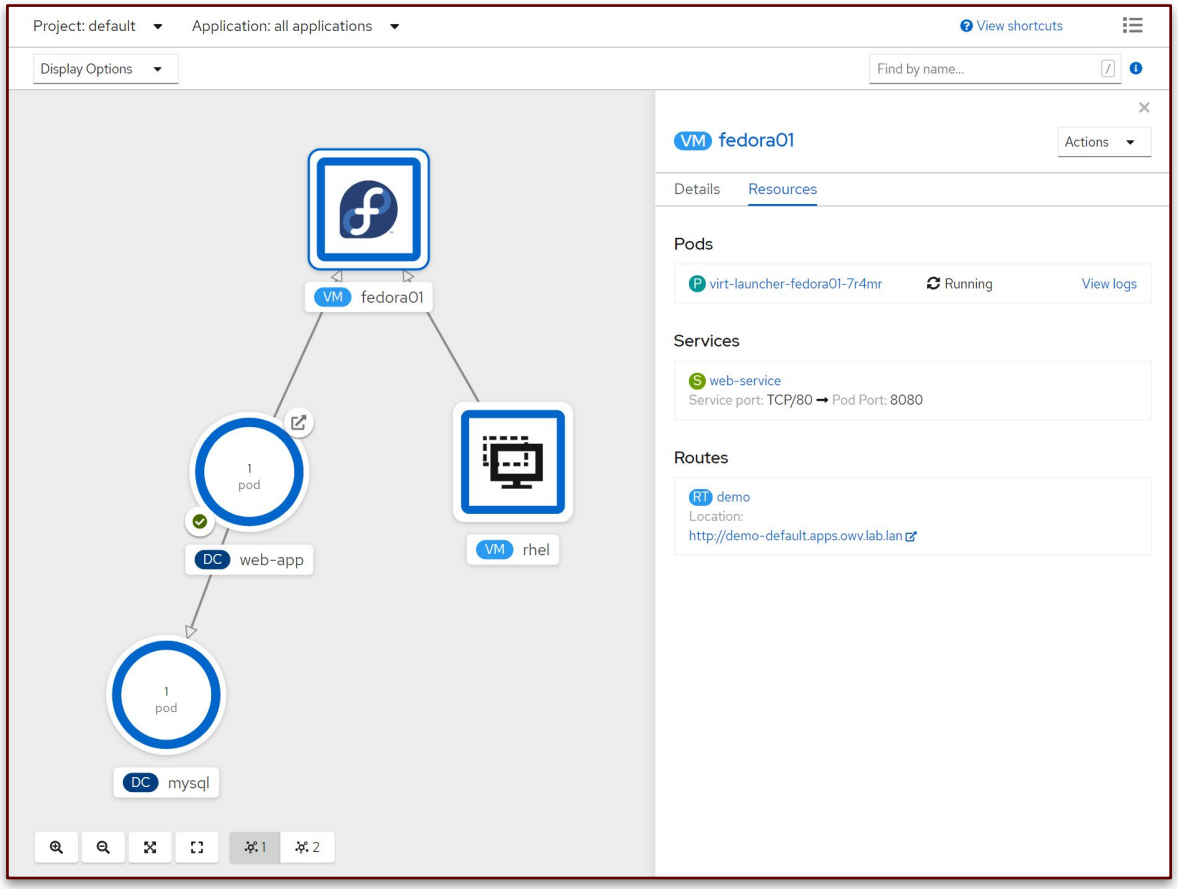
Red Hat OpenStack Platform

Red Hat Virtualization

Red Hat OpenShift

Using VMs and Containers Together

- Virtual machines connected to pod networks are accessible using standard Kubernetes methods:
  - Service
  - Route
  - Ingress
- Network policies apply to VM pods the same as application pods
- VM-to-pod, and vice-versa, communication happens over SDN or ingress depending on network connectivity



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Why organizations care

- Modernize without rewriting every legacy application
- Reduce silos between VM and container teams
- Faster delivery with shared CI/CD and GitOps
- Consistent security, compliance, and observability

Classic stacks vs OpenShift

- Separate virtualization and container platforms
- Different tooling, teams, and runbooks
- Harder unified governance and chargeback
- Slower handoffs between VM and app teams

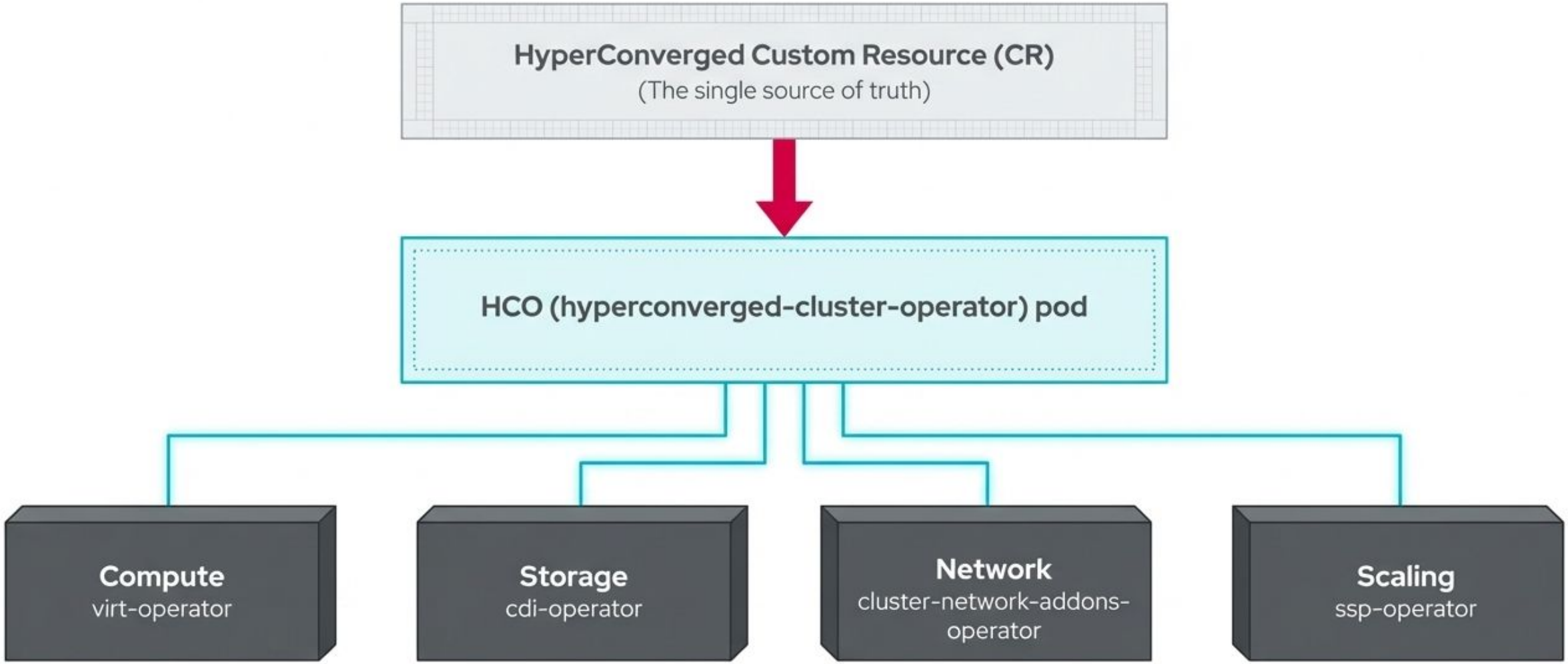
- Single Kubernetes control plane for VMs and pods
- Shared networking, storage, and identity models
- One observability and policy story end to end
- Gradual modernization with less rip-and-replace

Technical foundation

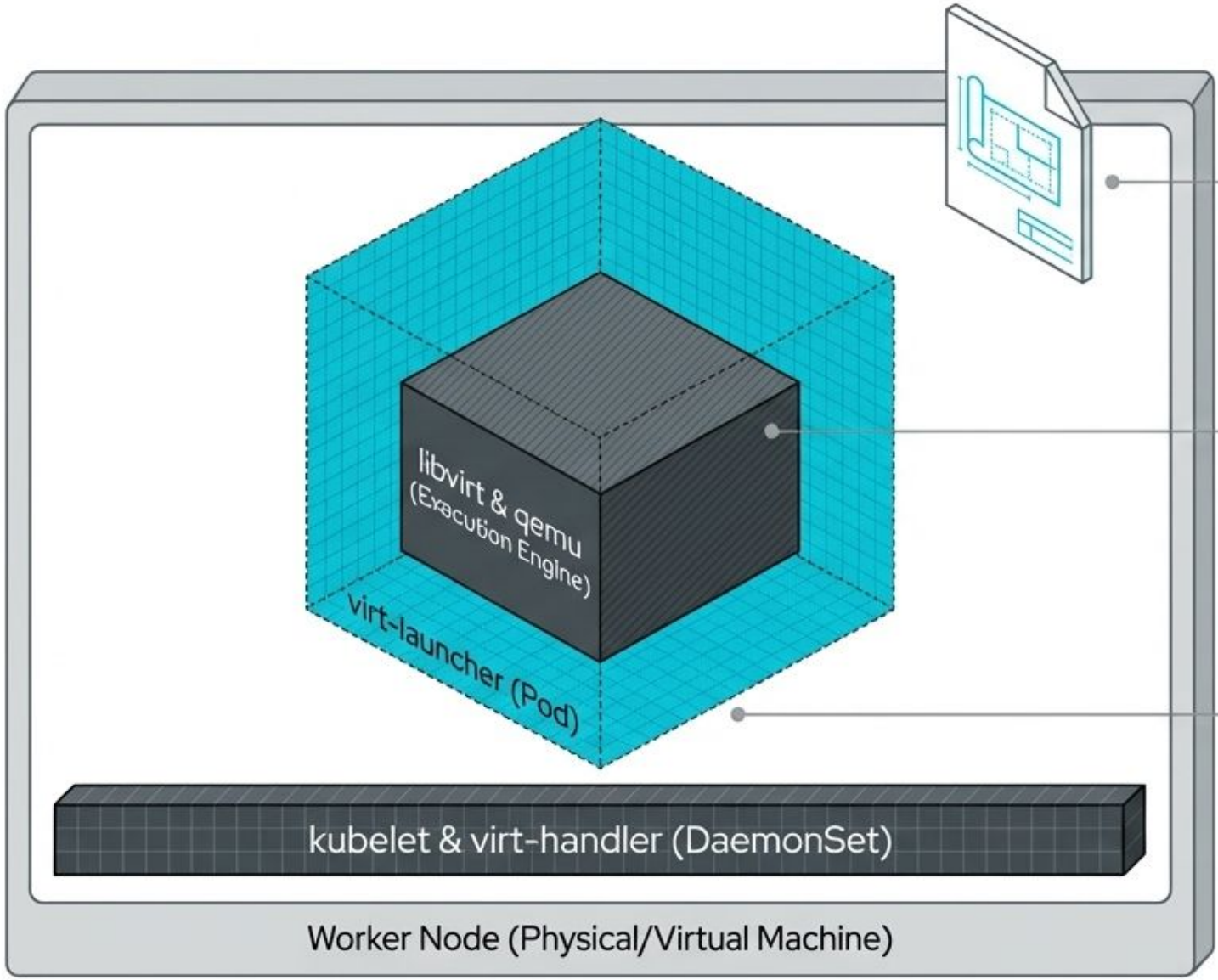
How it works (high level)

- VM specs as Kubernetes custom resources
- KubeVirt controllers schedule VM pods on nodes
- libvirt and QEMU run the guest workloads
- Integrated with OpenShift networking and storage
- Live migration for planned maintenance
- Templates and golden images for repeatability
- Monitoring and metrics via OpenShift observability
- Optional GPU, SR-IOV, and high-performance paths

The Control Plane: HyperConverged Operator (HCO)



The Execution Engine: Inside the virt-launcher



VirtualMachine (VM)
The declarative CRD describing the desired state.

VirtualMachineInstance (VMI)
The active representation of the running VM guest OS.

virt-launcher
The container holding the VM, mapping K8s lifecycle events (like SIGTERM) to libvirt domain events.

Networking at a glance

- Multus and SDN integrate VMs like other workloads
- Overlay and secondary networks for isolation or performance
- Services and routes expose apps consistently
- DNS and policy align with cluster networking guardrails

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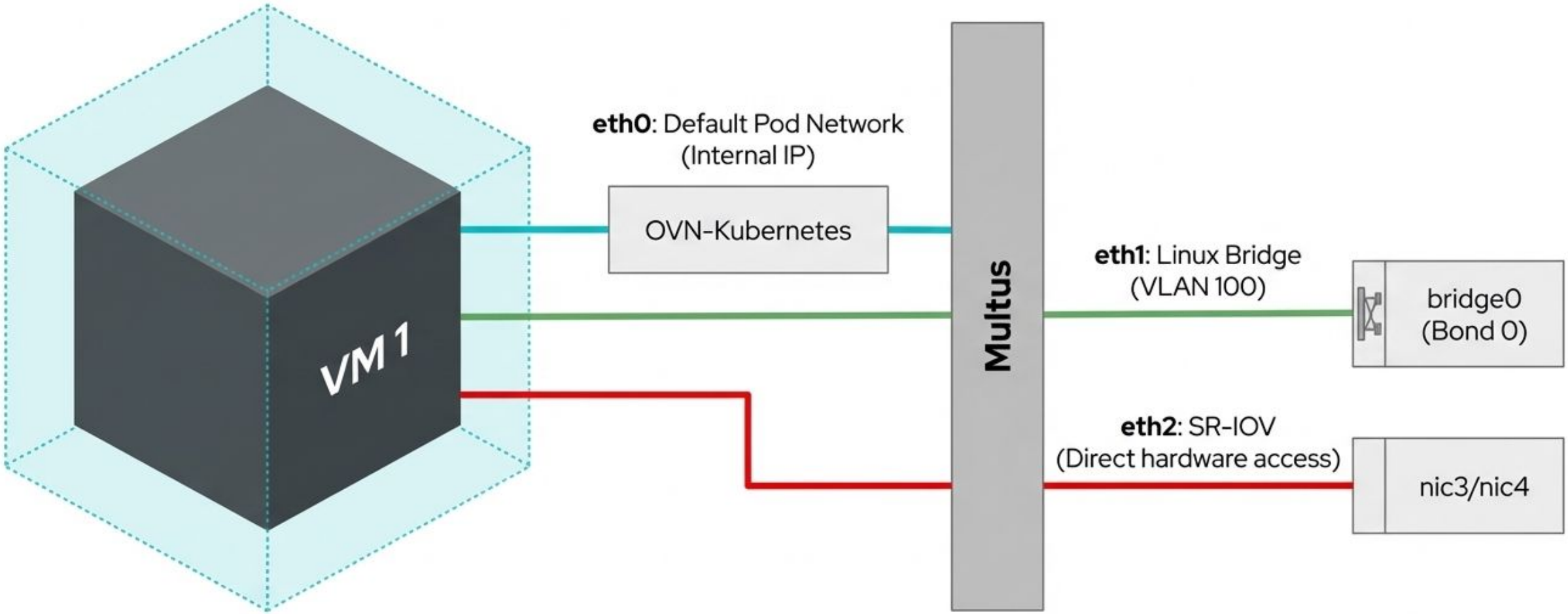
Secondary Networks: The Multus Meta-Plugin

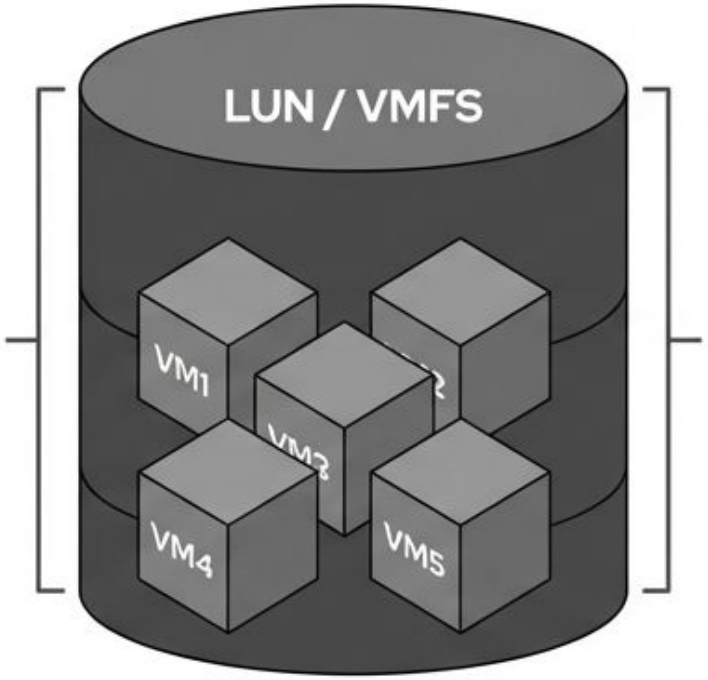


Storage and images

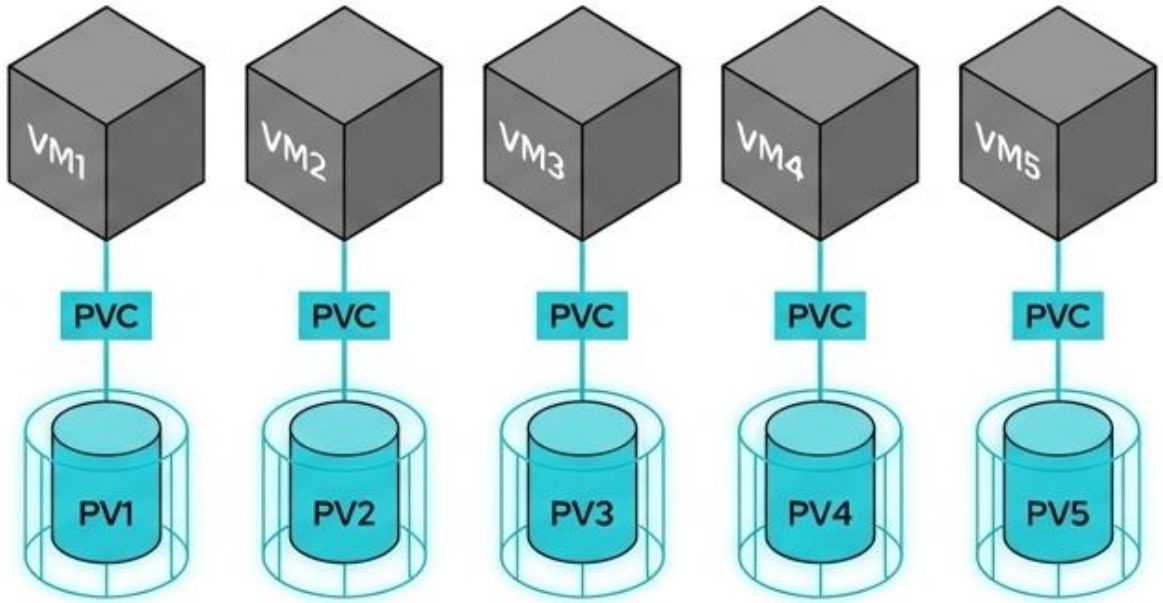
- PVC-backed disks with supported CSI drivers
- Golden images and templates from registries or HTTP
- Snapshots and clones for test and rollback scenarios
- Size performance tiers to workload classes early

Storage Transformation: Retiring the Datastore

vSphere Datastore



CSI & Persistent Volumes



Standardization

VM disks are managed exactly like container persistent storage.

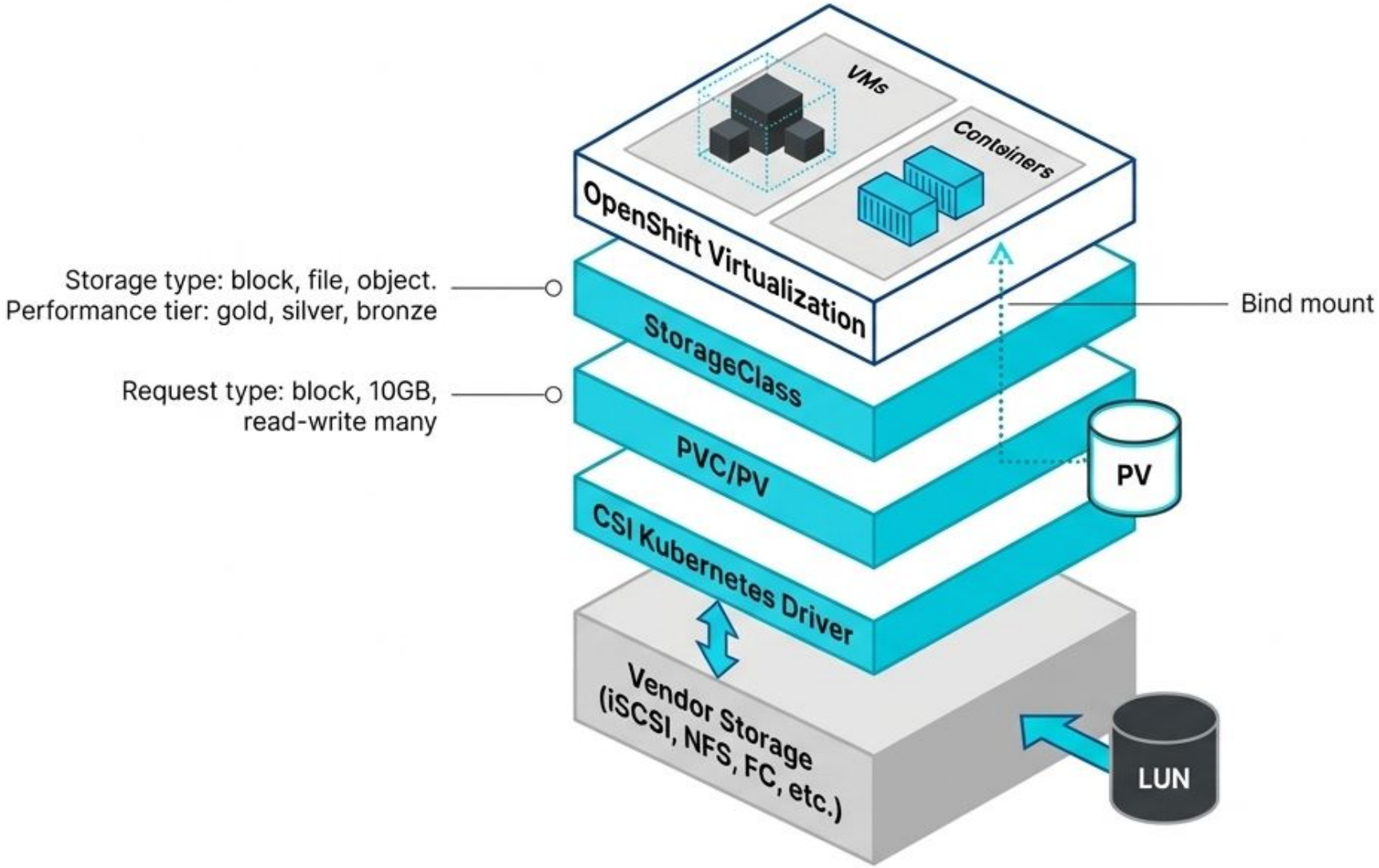
Dynamic Provisioning

StorageClasses automate volume provisioning based on performance tiers (gold, silver, bronze).

Access Modes

Seamless support for ReadWriteOnce (RWO) for standalone VMs, and ReadWriteMany (RWX) for live migration.

The Storage Flow:
From Vendor LUN to Virtual Machine



Migration and import

- Migration Toolkit for Virtualization for bulk moves
- Cold migration and cutover windows reduce risk
- Inventory dependencies: DNS, AD, licenses, and firewalls
- Pilot waves before wide production conversion

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Migration Toolkit for Virtualization (MTV)

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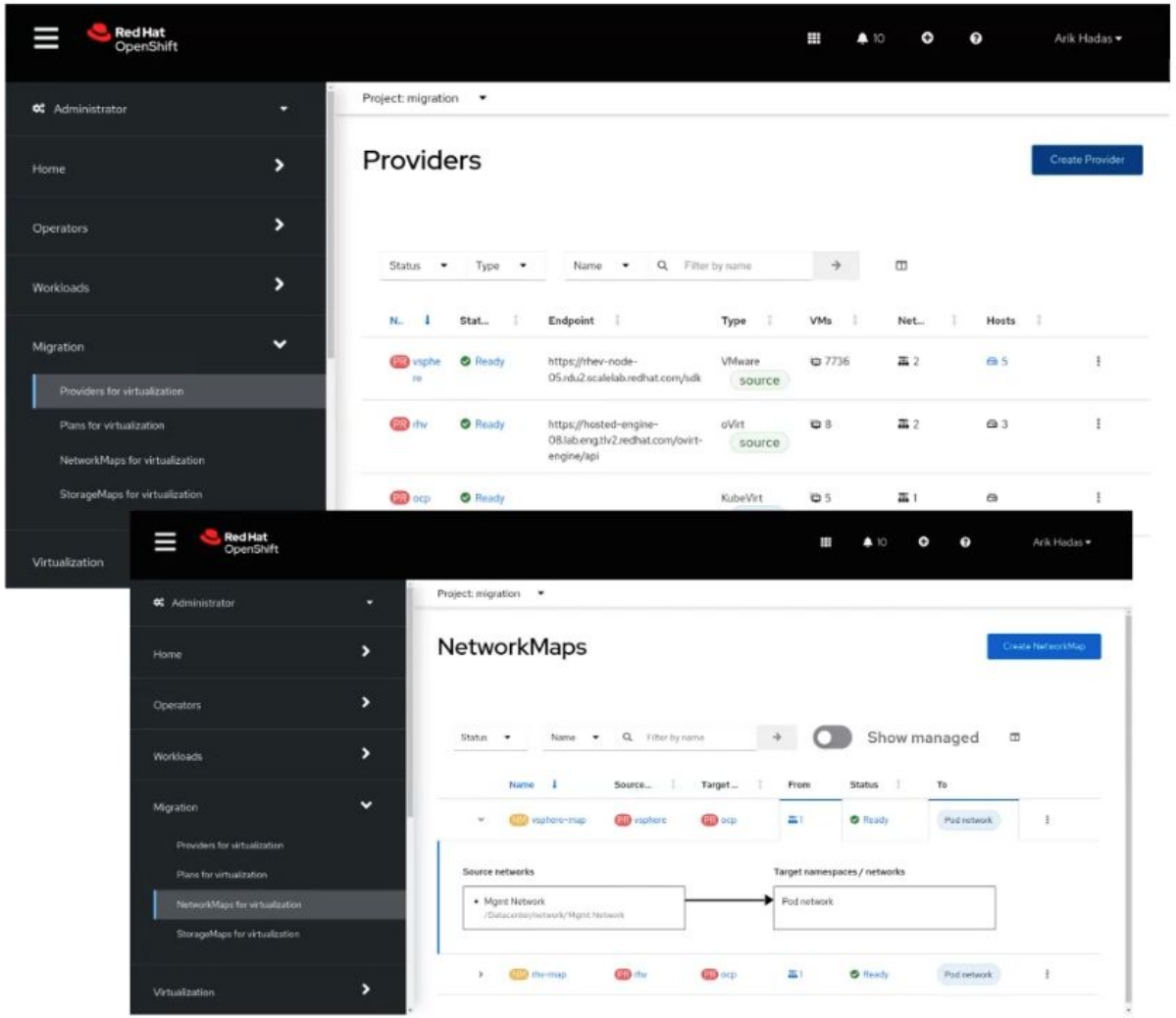
Main Features:

- Easy to use UI
- Mass migration of VMs from VMware, Red Hat Virtualization, OpenStack to OpenShift and between OpenShift Clusters
- VM data pre-copied before shutdown (Warm Migration) for VMware and Red Hat Virtualization migrations
- VM validation service:
  - Runs checks on VM configuration to avoid migration issues
- Parallelized VM Conversion
  - Maximize Throughput
- Migration Network Selection
  - Avoid impact on other running workloads



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Modernize at your own pace

Legacy Virtualization
Apps in VMs

Slow evolution

Increasing costs

Developer toil

Migration Toolkit for VMs

Infrastructure Modernization
Apps in VMs

Cloud elasticity + scalability

Reduced cost

Increase IT efficiency + reliability

Cloud Native

DevOps & Infrastructure Modernization
Apps in VMs or Containers

Innovate at speed

Higher annual revenue

Increased developer output

Direct path to cloud native

Speed of Infrastructure Deployment
Speed of Application Development

Operations and trust

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Security and compliance

- RBAC, SCCs, and namespaces segment who can do what
- Audit logs and compliance frameworks align with OCP
- Scan images and enforce policy with OpenShift tools
- Secrets and certificates integrate with existing IdM

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GitOps and automation

- VMs and data volumes as versioned manifests
- Pipelines promote changes like any app artifact
- Argo CD or OpenShift GitOps for drift control
- Repeatable environments from dev through production

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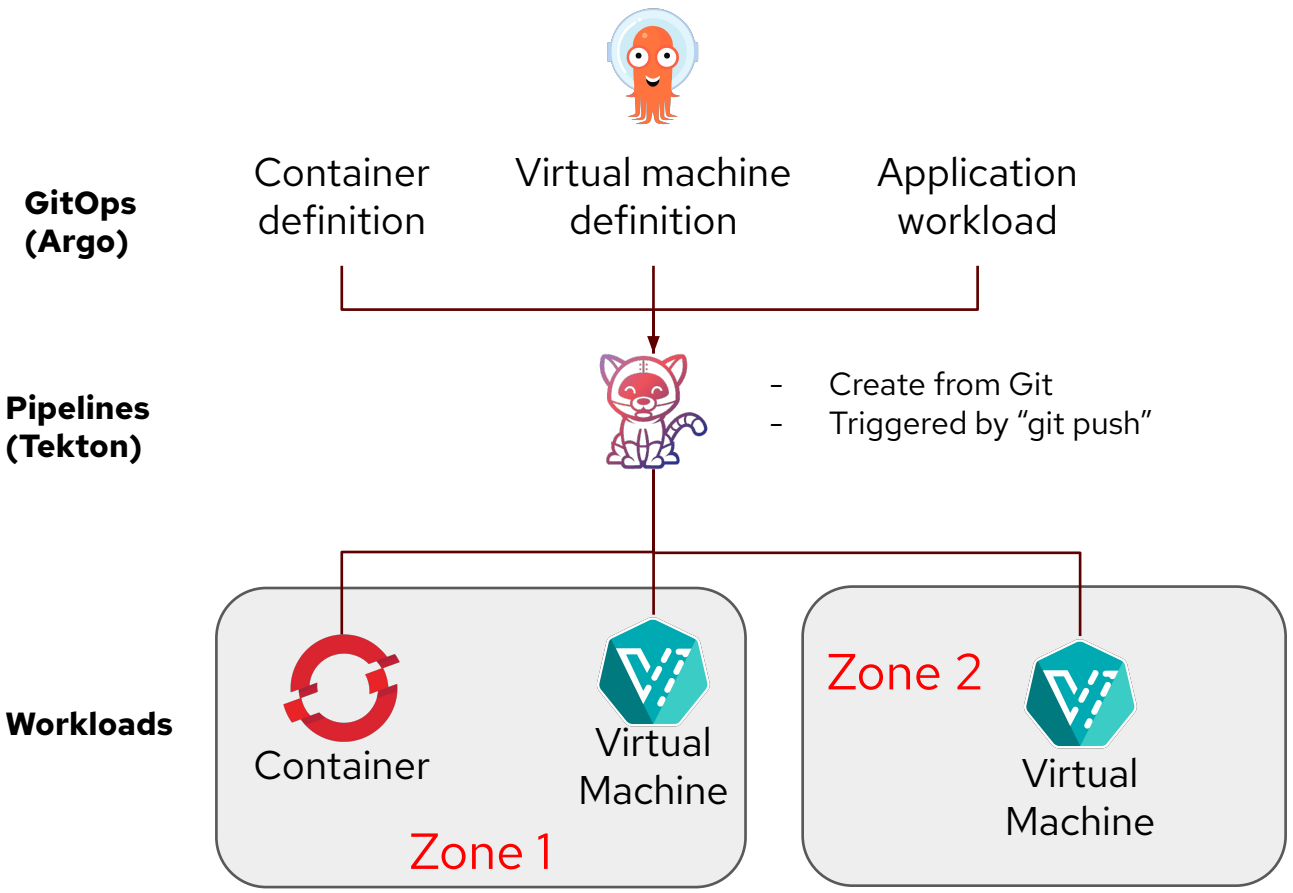
OpenShift Virtualization: Build Cloud-native VMs

Deploy VMs as Code with CI/CD



Integrate legacy VMs with a modern GitOps framework

- Deploy different security zones to run both composite applications of pods/VMs as well as traditional VM workloads
- Deploy and automate Virtual Machines as Code with GitOps

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Day-two operations

- Capacity planning for CPU, memory, and oversubscription
- Upgrade coordination with OpenShift release cadence
- Backup and disaster recovery with supported patterns
- Troubleshooting via console, CLI, and must-gather

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Adoption

Common use cases

- Lift-and-shift or refactor-later Windows and Linux VMs
- Mixed workloads during cloud or datacenter migration
- Dev and test environments matching production VMs
- Edge or remote sites needing local virtual machines

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Key takeaways

- OpenShift Virtualization unifies VMs with Kubernetes ops
- Plan network, storage, and migration before scale-out
- Reuse GitOps, security, and observability you already trust
- Start with a pilot wave; expand with proven runbooks

One Platform. One API. All Workloads.

Unified Control Plane

Eliminate the operational divide. Manage hypervisors and containers with a single skillset and declarative architecture.

Enterprise Resilience

Zero compromises. Full support for Live Migration, SR-IOV bare-bare-metal networking, and zero-copy CSI storage cloning.

Future-Proof Infrastructure

Modernize legacy stateful VMs at your own pace while building cloud-native microservices on the exact same physical nodes.

Thank you

Questions?

OpenShift Virtualization documentation |
https://docs.redhat.com/en/documentation/openshift_container_platform/

linkedin.com/company/red-hat

youtube.com/user/RedHatVideos

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